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SwiftLog Logistics Analytics

Data Analytics Track | Part 5: Business Analysis and Data Storytelling and Part 6: Business Recommendations

Key Takeaways

- **Delivery is the biggest problem:** only 36.28% of shipments arrive on time against an assumed 80% target, and no warehouse or city is close to it (the best warehouse, W01, reaches 41.3%).
- **Only a third of orders are delivered:** 33.2% of orders are Delivered, 32.5% are Cancelled and 34.3% are Pending, yet every order has a shipment. Delivered revenue is **₦7.11M** out of **₦21.67M** of total order value.
- **Stock is far larger than demand:** 1,007,764 units are held against 82,442 units ordered (12.2 times), while 160 products still have a warehouse below 100 units. This points to distribution, not total quantity, as the problem.
- **Revenue and customers are spread out:** the top 10 customers generate only 5.05% of delivered revenue and the top 10 products 9.85%, so growth depends on the whole base.
- **Returns are a process issue:** return rates are almost identical across categories (9.4% to 9.6%), and 63% of returns are Wrong Item or Damaged.
- **Suppliers cannot be assessed from the data:** there is no supplier link, so this report recommends data fixes instead of unsupported supplier rankings.

Basis of the Figures

- **Revenue** = Quantity × UnitPrice for Delivered orders (₦7,107,471). Cancelled (₦7,228,848) and Pending (₦7,331,748) order values are reported separately. Currency is not stated in the data; the naira sign is used as the reporting currency.
- **On Time** = 0 to 2 days of delay; **Delayed** = 3 to 7 days. The 80% on-time target is an assumed management target (a What-If parameter in the dashboard).
- **Delivery KPIs** use all 5,000 shipments, because every order has a shipment whatever its status.
- **Returns** link to orders, not products, so product return rates attribute each return to every item in the returned order and are approximate.
- **Inventory** is the sum of stock movements per warehouse and product; estimated utilisation assumes 80,000 units of capacity per warehouse.
- **September 2026** covers only 1 to 8 September and is excluded from trend comparisons.
- **Suppliers** are a standalone table with no link to products or shipments (admin guidance).

Part 5: Business Analysis and Data Storytelling

Each question is answered with evidence from the dashboard. The Dashboard column names the page and visual that supports the answer.

S/N	Question	Answer	Evidence	Dashboard support	Business meaning
1	Which products generate the highest revenue?	P203 (Fashion, ₦85,978) leads, followed by P085 (Food, ₦83,844), P229 (Food, ₦79,261), P006 (Fashion, ₦72,848) and P159 (Fashion, ₦65,589). The top 10 products earn only 9.85% of delivered revenue.	Top 10 products are all priced between ₦416 and ₦495. Revenue by category: Fashion ₦2.73M (38.4%), Food ₦2.37M (33.3%), Electronics ₦2.01M (28.3%). The top 50 products (20%) earn 38.4% of revenue.	Supplier and Product: Revenue by Category; Revenue by Product and Units Ordered	Revenue is led by high-priced Fashion and Food items but is spread widely, so no single product creates dependency.
2	Which customers contribute the highest revenue?	C0223 (₦57,592), C0773 (₦52,252), C0599 (₦40,996), C0520 (₦32,240) and C0030 (₦31,818) are the top five. The top 10 customers generate 5.05% of delivered revenue.	The top 100 customers (10% of the base) generate 32.0% and the top 200 (20%) generate 52.5% of delivered revenue. Revenue by city: Kano ₦1.86M, Lagos ₦1.81M, Ibadan ₦1.76M, Abuja ₦1.68M.	Customers Analysis: Top 10 Customers; Total Revenue by CustomerCity	No individual customer is critical, but the top fifth of customers carries about half of revenue and deserves a retention programme.
3	Which warehouses operate most efficiently?	Efficiency is judged by on-time rate and average delay. Most efficient: W01 Abuja (41.3%, 3.46 days), W10 Kano (40.1%, 3.49) and W07 Lagos (39.6%, 3.47). Least efficient: W05 (32.0%, 3.79 days), W09 (32.5%) and W03 (32.6%).	No warehouse is near the 80% target; the gap between best and worst is 9 points. W14 (9 staff, 37.8 shipments per employee) and W15 (35.2) have the heaviest load but are at or above the 36.3% average (38.5% and 36.4%). Workload and stock level show almost no link to on-time rate (correlations 0.13 and -0.07).	Inventory and Warehouse: Warehouse Comparison Table; Shipment and Delivery: On-Time Rate by WarehouseCity	Delays are systemic, not caused by one overloaded site. Good practice from W01, W10 and W07 should be copied to W05, W09 and W03.
4	Which suppliers consistently perform well?	This cannot be determined from the data. The Suppliers file has only SupplierID and SupplierName (150 suppliers) and nothing links a supplier to a product or shipment.	A network-level Supplier Performance Score is 58.3 out of 100 (60% weight on on-time delivery, 40% on return-free orders), by category: Electronics 57.7, Fashion 58.4, Food 58.8. Admin confirmed that suppliers are a standalone reference table.	Supplier and Product: Supplier cards and Supplier Score by Category (with assumption note)	Supplier decisions should wait until a supplier ID is added to products. The score is a framework, not a ranking.

S/N	Question	Answer	Evidence	Dashboard support	Business meaning
5	Which products have the highest return rates?	Overall return rate is 9.46% (473 of 5,000 orders). Among high-volume products (75+ order lines, shown on the dashboard): P130 13.9%, P007 13.2% and P140 12.7%. Across all 250 products the highest rates are P032 (21.3%), P017 (18.9%) and P108 (18.3%), but on small samples of 47 to 60 lines.	Return rate by category is almost the same: Electronics 9.6%, Food 9.4%, Fashion 9.4%. Return reasons: Late Delivery 185, Wrong Item 163, Damaged 152. With 250 products, some high rates are expected by chance.	Supplier and Product: Returned Product Rate; Executive Overview: Return Overview	Returns are a company-wide process issue, not a product-specific one. Track products with high rates over more months before acting.
6	Which regions experience the most delayed deliveries?	By customer city, Kano is worst: 65.7% of shipments delayed (845 shipments, 3.64 days average). Lagos is best at 61.0%. By warehouse city, Lagos warehouses have the most delays: 1,701 of 2,647 shipments (64.3%). Kano warehouses are best at 62.5%.	On-time rate by customer city: Lagos 39.0%, Ibadan 36.2%, Abuja 35.6%, Kano 34.3%. On-time rate by warehouse city: Kano 37.5%, Abuja 36.4%, Lagos 35.7%. Lagos warehouses ship 53% of all shipments.	Shipment and Delivery: On-Time Delivery Rate by CustomerCity and by WarehouseCity	The gap between regions is only 4 to 5 points, so delays occur everywhere. Lagos matters most because of its volume.
7	Which months record peak business activity?	August 2026 is the peak: delivered revenue ₦997,021 and 654 orders (the highest of any month). Weakest revenue month: July (₦741,706).	August also has 221 delivered orders and the lowest cancellation rate (29.2%). Second peaks: April for orders (638) and June for revenue (₦914,799). Monthly orders stay between 577 and 654, so demand is fairly flat. September covers 8 days only (154 orders).	Executive Overview: Revenue by Month; Monthly Orders	Demand is steady with an August peak. Capacity and stock plans should prepare for peak months.

S/N	Question	Answer	Evidence	Dashboard support	Business meaning
8	Which KPIs require immediate management attention?	1. On-Time Delivery Rate 36.28% against an assumed 80% target. 2. Cancellation Rate 32.52% (₦7.23M of order value) and Pending orders 34.32% (₦7.33M). 3. Inventory Turnover 0.027. 4. Return Rate 9.46%.	Average delay is 3.57 days (5.03 days among delayed shipments); 1,965 shipments (39%) are 5 to 7 days late. Only 33.2% of orders (1,658) are Delivered. Stock is 12.2 times the units ordered. Late Delivery is 37% of returns.	Executive Overview: KPI cards (conditional colours); Inventory and Warehouse: Inventory Turnover; Shipment and Delivery cards	Delivery speed and order completion drive most of the lost value, and should lead the management agenda.
9	What operational inefficiencies were identified?	1. Late deliveries: 63.7% of shipments delayed. 2. Shipping orders that are not delivered: all 1,626 Cancelled and 1,716 Pending orders have a shipment. 3. Overstock beside shortages: 12.2 times the stock needed, yet 160 products have a warehouse under 100 units. 4. Uneven staffing: 19.9 to 37.8 shipments per employee. 5. Cross-city shipping: only 24.9% of shipments come from the customer's own city; Ibadan has no warehouse. 6. Returns from handling errors: 63% are Wrong Item or Damaged.	315 of the 473 returned orders are Cancelled or Pending. Ibadan: 262 customers and ₦1.76M revenue, with 52% of its shipments sent from Lagos. 236 orders (4.7%) have no items. Late Delivery returns are no more delayed than others (35.7% on time vs 37.1%), so the reason code may be unreliable.	Shipment and Delivery; Inventory and Warehouse; Executive Overview (Return Overview)	The same data shows both excess and shortage and unreliable order status, so process control, not capacity, is the main issue.

S/N	Question	Answer	Evidence	Dashboard support	Business meaning
10	What business opportunities should SwiftLog pursue?	1. Recover cancelled and pending orders. Order value is ₦21.67M but only ₦7.11M is delivered. 2. Improve on-time delivery toward the target. 3. Release money tied up in stock. 4. Serve Ibadan better. 5. Grow Fashion and high-priced products.	Each 10 points off the cancellation rate is about 500 more delivered orders, roughly ₦2.2M at the ₦4,504 average order value. Each 10 points on on-time rate is about 500 more on-time shipments. Ibadan is 26% of customers and 25% of revenue. Fashion is 38.4% of revenue and top products sell at ₦416 to ₦495.	Executive Overview; Customers Analysis; Supplier and Product: Revenue by Category	The largest upside comes from fixing order completion and delivery, which use existing customers and stock.

Limitations

- **No delivery timestamps:** delivery time cannot be measured directly; only the delay in days is available.
- **No warehouse capacity:** utilization is estimated using an assumed capacity of 80,000 units.
- **No supplier link:** supplier performance is not rankable.
- **Returns have no product:** product return rates are approximate, and small samples can show high rates by chance.
- **Order status conflicts with shipments:** Cancelled and Pending orders have shipments and returns, so status quality needs checking.
- **No cost data:** impact figures use revenue only and are illustrative, not profit forecasts.
- **September is partial,** and the data covers under nine months.

Part 6: Business Recommendations

Format: Finding → Evidence → Recommended Action → Expected Impact. Every recommendation rests on a finding from the dashboard. Impact figures are illustrative arithmetic on revenue, not profit forecasts.

S/N	Area	Finding	Evidence	Recommended action	Expected impact and KPI	Dashboard support
R1	Delivery optimisation	On-time delivery is far below target.	On-time rate 36.28%; 3,186 shipments delayed; 1,965 are 5 to 7 days late; average delay 3.57 days.	Set an interim target of 50% on-time in 90 days. Flag every shipment late by more than 2 days for escalation, and review delayed shipments weekly with the carriers.	Each +10 points = about 500 more on-time shipments. Track On-Time Delivery Rate.	Shipment and Delivery

S/N	Area	Finding	Evidence	Recommended action	Expected impact and KPI	Dashboard support
R2	Delivery optimisation and warehouse efficiency	Warehouse results differ by 9 points.	W01 41.3%, W10 40.1%, W07 39.6% against W05 32.0%, W09 32.5%, W03 32.6%.	Document the dispatch process used at W01, W10 and W07 and apply it at W05, W09 and W03. Review again after 60 days.	Lifting the three weakest to the average (36.3%) adds about 40 on-time shipments per 1,000 handled.	Inventory and Warehouse; Shipment and Delivery
R3	Warehouse efficiency	Workload per employee varies almost twofold.	Shipments per employee: W14 37.8, W15 35.2, W12 33.7 against W13 19.9 and W03 20.2.	Rebalance staff or route shipments so no warehouse exceeds about 30 per employee. Keep the change cost-neutral by moving staff from lighter sites.	More even workload and lower overtime risk. Track Shipments per Employee.	Inventory and Warehouse
R4	Inventory management	Total stock far exceeds demand.	1,007,764 units in stock against 82,442 ordered (12.2 times); turnover 0.027.	Set maximum stock levels and reorder points by product, run a stock count to confirm the movement records, and slow replenishment of overstocked lines.	Frees working capital and storage space. Track Inventory Turnover.	Inventory and Warehouse
R5	Inventory management	Shortages exist beside overstock.	160 products have at least one warehouse below 100 units (264 lines); average stock per warehouse and product is about 410 units.	Move stock from overstocked to short warehouses and use a minimum-stock rule by warehouse and product, reviewed weekly with the Low Stock table.	Fewer stock-outs without buying more. Track Low Stock Products.	Inventory and Warehouse
R6	Operational cost reduction	Orders that are not delivered still get shipments.	1,626 Cancelled and 1,716 Pending orders (66.8%) all have shipments; their value is ₦7.23M and ₦7.33M.	Do not dispatch until an order is confirmed, clear the pending backlog, and investigate cancellation reasons. Fix the order status so it matches the shipment status.	Up to 1,626 avoided shipments. Cutting cancellations by 10 points adds about 500 delivered orders (₦2.2M).	Executive Overview

S/N	Area	Finding	Evidence	Recommended action	Expected impact and KPI	Dashboard support
R7	Customer retention	Revenue relies on the top fifth of customers.	Top 200 customers (20%) generate 52.5% of delivered revenue; 132 customers have only 1 or 2 orders; 9 never ordered; retention rate 95.96%.	Offer a loyalty or priority-service programme to the top 200 customers and a win-back offer to the 132 low-frequency customers.	Protects over half of delivered revenue. Track Customer Retention Rate and Orders per Customer.	Customers Analysis
R8	Return reduction	Most returns come from handling errors.	500 returns (473 orders, 9.46%): Late Delivery 185, Wrong Item 163, Damaged 152; returned delivered orders carry ₦715,766 (10.1% of delivered revenue).	Add pick-and-pack accuracy checks and stronger packaging to cut Wrong Item and Damaged returns, and check that the Late Delivery reason matches real delays.	Targeting the 315 Wrong Item and Damaged returns; return rate target of 7% (proposed).	Executive Overview; Supplier and Product
R9	Supplier improvement	Supplier performance cannot be measured.	150 suppliers; no field links suppliers to products or shipments; network score 58.3 out of 100.	Add SupplierID to Products and Shipments, then score suppliers on on-time delivery, return and defect rate, and lead time. Use the dashboard score weights as the starting framework and include them in supplier contracts.	Enables real supplier rankings and service agreements. Track Supplier Performance Score.	Supplier and Product
R10	Logistics optimisation	Shipments rarely leave from the customer's own city.	Only 24.9% of shipments come from the customer's city; Ibadan (262 customers, ₦1.76M) has no warehouse, and 644 of its 1,240 shipments leave from Lagos.	Pilot nearest-warehouse rules for Ibadan and Kano, and assess an Ibadan hub. Measure the result first, because same-city and cross-city on-time rates are currently similar (63.9% and 63.7% delayed).	Shorter routes and lower transport cost if the pilot succeeds. Track On-Time Rate by city.	Shipment and Delivery; Customers Analysis

S/N	Area	Finding	Evidence	Recommended action	Expected impact and KPI	Dashboard support
R11	Performance monitoring	No targets or timestamps exist to judge performance.	Dispatch and delivery dates, warehouse capacity and supplier links are missing; order status conflicts with shipments.	Use the dashboard as a weekly KPI scorecard with targets: on-time 80%, cancellation 20% or less, return rate 7% or less, rising turnover. Add dispatch and delivery timestamps and warehouse capacity to the data.	Measurable accountability for each KPI owner.	All pages (slicers, conditional formatting)
R12	Business growth opportunities	Revenue is strongest in a few high-priced categories and in August.	Fashion ₦2.73M (38.4%); top 10 products priced ₦416 to ₦495; August revenue ₦997,021 and 654 orders.	Expand the high-priced Fashion and Food assortment, run promotions in the weaker months (July revenue ₦741,706), and plan stock and staff ahead of the August peak.	More revenue from the existing customer base. Track Revenue by Month and by Category.	Executive Overview; Supplier and Product